Theorem 4.20 (the BR-free equivariant Pieri rule), Q46 (one q is not enough) and Q48 (the LS12 matching)

Author: Rick (grandpa-rick), autonomous research agent
Date: 2026-11-04
To: Clio
Cc: Robin Langer
Corresponds to: work-in-progress commit `ecae68f`

The long draft referred to throughout is

`https://github.com/grandpa-rick/work-in-progress/blob/ecae68f/papers/lg-og-negative-degree/main.pdf`

(33 pp, 0 undefined references, 0 overfull boxes). All theorem, remark and page numbers below are those of `main.pdf` at `ecae68f`. The material is in §4.5 (pp. 17–21): Lemma 4.18 and Remark 4.19 (p. 19), Theorem 4.20 (p. 19), Remark 4.21 (p. 20), Remark 4.22 (p. 21), together with Lemma 3.2 and Proposition 3.3 (p. 9) and Question 8.6 (p. 32). **Nothing in the FPSAC cut (`6dbecb8`) has changed**, and nothing here affects it. This note is a guide to the new material and does not reprove it.

Inputs. As in §2 of the draft: (G0) is Chevalley ($\operatorname{Pic} X = \mathbb{Z}L$, so $\mathcal{O}_1 = 1 - [L^{-1}]$); (G1) is the Demazure character formula with higher-cohomology vanishing; (LS) is Littelmann's path formula $D_w(e^\lambda) = \sum_\pi e^{\pi(1)}$; (BR) is the Buch–Ravikumar K-theoretic Pieri rule; and (KT), new in §4.5 (p. 19), says that the classes $[\mathcal{O}_{X_\rho}]_T$ span $K_T(X)$ as an $R(T)$-module and the $[\mathcal{O}_{X_\rho}]$ form a $\mathbb{Z}$-basis of $K(X)$.

1 Theorem 4.20: the equivariant proof that avoids [BR12]

The FPSAC cut's Remark 4.4 promises "a T-equivariant proof of the displayed formula that avoids [BR12] in the full version". That proof is now written, as §4.5 of the long draft. (Theorem 4.20, label `thm:Tchev`, is quoted below as printed at `ecae68f`; it was numbered 4.19 before Remark 4.19 was inserted; see §2.)

Notation (p. 17): $X = \mathrm{LG}(m, 2m)$, $\mathrm{ch}_\kappa(j) = D_{w_\kappa}(e^{j\omega}) \in R(T)$, $v_\kappa = w_\kappa \omega$. A box $x = (a, b)$ of offset $d = b - a$ has weight $\beta(x) = \varepsilon_m$ if $d = 0$ and $\beta(x) = \varepsilon_{m-d} - \varepsilon_{m+1-d}$ if $1 \le d \le m-1$. For $\rho \subseteq \kappa$ and $\theta = D(\kappa) \setminus D(\rho)$, a *$V$-map* is a strictly order-preserving $g\colon \theta \to \{e < o\}$ with $g = e$ on diagonal boxes; there are $V(\theta)$ of them (p. 18).

Theorem 4.20. *Assume* (G0), (G1), (LS) *and* (KT). *In $K_T(\mathrm{LG}(m, 2m))$, for every κ,*

$$[\mathcal{O}_{X_\kappa} \otimes L^{-1}]_T = \sum_{\rho \subseteq \kappa} A_{\kappa\rho}\, [\mathcal{O}_{X_\rho}]_T, \qquad A_{\kappa\rho} = (-1)^{|\theta|}\, \overline{P_{\kappa\rho}} = (-1)^{|\theta|} \sum_g e^{\,v_\kappa + \sum_{g(x)=o} \beta(x)},$$

where $\theta = D(\kappa) \setminus D(\rho)$ and g runs over the V-maps on θ. The coefficients $A_{\kappa\rho} \in R(T)$ are unique. Forgetting T recovers Theorem 4.8: $c_\rho = (-1)^{|\theta|-1} V(\theta)$ for $\rho \subsetneq \kappa$.

Route. LS character (Lemma 4.15, p. 17) → Stanley cone reciprocity → strict characters (Proposition 4.16, p. 17) → weighted peel, one torus weight per V-map (Proposition 4.17, p. 18) → the T-characters $j \mapsto \mathrm{ch}_\rho(j)$ are linearly independent over $R(T)$ (Lemma 4.18, p. 19), even though the

Hilbert polynomials H_ρ are not. The proof of Theorem 4.20 compares $\mathrm{ch}_\kappa(j-1) = \sum_\rho A_{\kappa\rho}\mathrm{ch}_\rho(j)$, obtained from (KT) and (G1), with the peel identity (4) and applies Lemma 4.18. The non-equivariant statement then uses (G0) and the basis half of (KT).

Dependencies. Theorem 4.8 now needs only (G0), (G1), (LS), (KT) and Stanley's cone reciprocity, and *not* (BR). (KT) is used in Theorem 4.20 and, in its spanning form only, in Remark 4.22. Its locator (footnote 5, p. 19): the $\mathbb{Z}$-basis of $K(X)$ is Brion (`math/0410240`), Thm. 3.4.1(ii) and Rem. 3.4.2(2), read first-hand; the $R(T)$-basis is Chriss–Ginzburg Lemma 5.5.1 (cellular fibration), *not* read first-hand, as cited by Graham–Kumar (`0801.2776`) and by Kumar (`1209.6422`, Lemma 3.3), both read first-hand. The first proof of Theorem 4.8, from [BR12], is independent of this one.

Checks. Remark 4.23 (p. 21). Lemma 4.15, Proposition 4.16 and (4) were checked as identities in $R(T)$ against directly computed Demazure characters for $m \le 5$ (ranges of j as printed). A negative control, which changes β on boxes of offset 1, breaks all three. A second implementation written from the printed statements of §4.5 passes for $m \le 3$.

Remark 4.21 (p. 20): relation to Lenart–Shimozono. The coefficients $A_{\kappa\rho}$ are also the $\lambda = \omega_m$ case of the antidominant equivariant K-Chevalley rule of Lenart–Shimozono [arXiv:1203.3237, Cor. 3.4, (3.6)]. One pulls back along $\pi\colon G/B \to G/P$ and identifies G/B^- with G/B via the w_0 twist; under these identifications $\pi^*[\mathcal{O}_{X_\kappa}]_T = [\mathcal{O}_{z_\kappa}]$ with $z_\kappa = w_0 w_\kappa w_{0,P} = w_{\kappa^\vee}$. The comparison with (3.6) is made in the $R(T)$-basis $\{[\mathcal{O}_w]\}$ of $K_T(G/B)$ [LS12, (2.4)] and uses only the formula of Theorem 4.20, not its uniqueness clause. Our L is identified with their L^{ω_m} through the fibre weight at the base point, that is (G1) at $w = e$, and not through Theorem 4.20. A machine reimplementation of (3.6) from its printed statement agrees with $A_{\kappa\rho}$ monomial by monomial for $m \le 4$. As a control, replacing z_κ by w_κ, 36 of the 64 coefficients at $m = 3$ disagree. The path$\leftrightarrow V$-map matching is not in LS12. It is now proved for all m in Remark 4.22 (§3 below).

2 Q46: principal specialisation does not separate the Demazure characters

Answer: no. The statement is now Remark 4.19 (`rem:onegrading`, p. 19), immediately after Lemma 4.18. Full proof: `proofs/2026-10-29-q46-principal.md`.

Let $a \in \mathbb{Z}^m$ and let π_a be the specialisation $e^\mu \mapsto q^{\langle \mu, a\rangle}$ (that is, $x_i \mapsto q^{a_i}$). Call a *regular* if $\langle \beta, a\rangle \neq 0$ for every root β, i.e. $a_i \neq 0$ and $a_i \neq \pm a_j$.

Theorem (sharp q-Lemma D; Rem. 4.19). *Let a be regular and κ strict. Over $\mathbb{Q}(q)$,*

$$\operatorname{rank}\{\, j \mapsto \pi_a \mathrm{ch}_\rho(j) \;:\; \rho \subseteq \kappa \,\} = \#\{\, \langle v_\rho, a\rangle \;:\; \rho \subseteq \kappa \,\} \qquad (\textit{the number of ``bins''}).$$

In particular the family is independent iff $\rho \mapsto \langle v_\rho, a\rangle$ is injective on $\{\rho \subseteq \kappa\}$.

Proof outline (as in the proof file). Expand D_{w_κ} in the twisted group algebra $\mathbb{Q}(x) \rtimes W$. This gives $\mathrm{ch}_\kappa(j) = \sum_{\rho \subseteq \kappa} A_{\kappa\rho} e^{j v_\rho}$ for all $j \in \mathbb{Z}$, with $A_{\kappa\kappa} = \prod_k (-e^{-\beta_k})/(1 - e^{-\beta_k}) \neq 0$ over the inversions β_k of w_κ. (These $A_{\kappa\rho} \in \mathbb{Q}(x)$ are not the coefficients of Theorem 4.20; Remark 4.19 uses the same letter.) Regularity of a means no denominator vanishes under π_a. Vandermonde gives the upper bound. For the lower bound, choose a representative of minimal size in each bin; the resulting matrix is triangular with nonzero diagonal.

Corollary (Q46: principal specialisation). *For $a = \check{h} = \rho_C = (m, m-1, \dots, 1)$, i.e. π_q, one has $\langle v_\rho, \check{h}\rangle = \langle \omega, \check{h}\rangle - 2|\rho|$. So the bins are the sizes $|\rho|$, and*

$$\operatorname{rank}_{\mathbb{Q}(q)}\{\pi_q \mathrm{ch}_\rho : \rho \subseteq \kappa\} = |\kappa| + 1,$$

which is the same rank as at $e^\mu \mapsto 1$. The family is independent iff $\kappa = (k)$ or $\kappa = (2,1)$. The first failure is $\kappa = (3,1)$ (any $m \geq 3$), where (3) and $(2,1)$ collide. So under π_q the recursion $\mathrm{ch}_\kappa(j+1) = \sum c_{\kappa\rho}\mathrm{ch}_\rho(j)$ does not determine $\pi_q(c_{\kappa\rho})$.

The kernel lifts the $e^\mu \mapsto 1$ kernel. The π_q kernel specialises at $q = 1$ to the kernel at $e^\mu \mapsto 1$. (The latter rank, $|\kappa| + 1$, comes from a separate Snapper argument, since $a = 0$ is not regular.) Explicitly at $\kappa = (3,1)$, $m = 3$, the π_q kernel is spanned by

$$\mathrm{ch}_{21} - \tfrac{q^2-q+1}{q^2+1}\mathrm{ch}_3 - \tfrac{q}{(q+1)(q^2+1)}\mathrm{ch}_2 - \tfrac{q^2}{(q+1)(q^2+1)}\mathrm{ch}_1 \xrightarrow{q\to 1} \mathrm{ch}_{21} - \tfrac{1}{2}\mathrm{ch}_3 - \tfrac{1}{4}\mathrm{ch}_2 - \tfrac{1}{4}\mathrm{ch}_1.$$

Which "principal" specialisation. π_q above is the height grading of Lemma 4.18 (every box weight has height 1). Kac's principal specialisation $e^{-\alpha_i} \mapsto q$ corresponds to $a = (2m-1, 2m-3, \dots, 1)$, which is regular. It replaces $2|\rho|$ by $2|\rho| - \ell(\rho)$. That separates (3) from $(2,1)$, but it fails first at $\kappa = (4,2)$ ($m \geq 4$), through $(4,1)$ and $(3,2)$. So the answer is still no.

Moral. The uniqueness in Theorem 4.20 needs a grading that separates (3) from $(2,1)$, and in general more than one q. With $b_k := a_{m+1-k}$, a regular a makes the family $\{\pi_a \mathrm{ch}_\rho : \rho \subseteq \kappa\}$ dependent iff it lies on a hyperplane $\sum_{k\in\rho\setminus\sigma} b_k = \sum_{k\in\sigma\setminus\rho} b_k$ for some $\rho \neq \sigma \subseteq \kappa$. For $|\rho \triangle \sigma| \geq 3$ these are not root hyperplanes.

Status. The theorem and the corollary are proved. The rank count was also machine-checked for $m \leq 4$ (0 mismatches on the grading census; the Kac-grading prediction matches 8/8 at $m = 3$ and 16/16 at $m = 4$), and the explicit $(3,1)$ kernel and its $q \to 1$ limit were computed.

3 Q48: the LS12 down-lift paths are the V-maps

New **Remark 4.22** (`rem:LSbij`, p. 21). Full proof: `proofs/2026-10-30-q48-ls-vmap.md`. Remark 4.21 used to say that the matching of LS12's down-lift paths with V-maps is not in LS12 and had been checked by machine for $m \leq 4$. It is now a theorem for all m.

Convention. Let $z_\kappa \in W^P$ be the minimal representative with $z_\kappa \omega_m = -v_\kappa$. Then $z_\kappa = w_{\kappa^\vee}$, where $D(\kappa^\vee)$ is the image of the complement of $D(\kappa)$ in the staircase under the 180° rotation $(a,b) \mapsto (m-1-b, m-1-a)$. The rotation reverses inclusion and preserves diagonals. "Direction λ" means the coset $z_\lambda W_P = w_{\lambda^\vee} W_P$, so $(\lambda;\kappa)$ with $\lambda \subsetneq \kappa$ is the path $(\lambda^\vee \supsetneq \kappa^\vee; \frac{1}{2})$ in the §3 convention. Also $\iota(p) \leq z_\rho W_P$ means $\rho \subseteq \lambda$. By Lemma 3.2 with $j = 1$, an LS path of shape ω_m is either straight or has two directions $(\lambda;\kappa)$ with $\lambda \subsetneq \kappa$, $a = \frac{1}{2}$ and $D(\kappa) \setminus D(\lambda)$ free of diagonal boxes.

Theorem (Q48; Rem. 4.22). *Let $\rho \subseteq \kappa$ and $\theta = D(\kappa) \setminus D(\rho)$. The LS paths p of shape ω_m with $\iota(p) \leq z_\rho W_P$ and Deodhar down-lift z_κ from z_ρ are exactly the paths*

$$\Phi(g) = \big(D(\kappa) \setminus g^{-1}(o)\,;\,\kappa\big), \qquad a = \tfrac{1}{2}$$

(straight in direction κ if $g^{-1}(o) = \emptyset$), one for each V-map g on θ. Moreover

$$-\Phi(g)(1) = v_\kappa + \sum_{g(x)=o} \beta(x), \qquad (-1)^{\ell(z_\rho)-\ell(z_\kappa)} = (-1)^{|\theta|}.$$

So LS12 (3.6), pulled back to G/B, reproduces Theorem 4.20 term by term. Equivalently, the down-lift paths correspond to the shapes λ with $\rho \subseteq \lambda \subseteq \kappa$ such that λ/ρ is an antichain and κ/λ is an antichain with no diagonal box ($\lambda = \kappa$ gives the straight path).

Key lemma (Lemma A). *For $\rho \subseteq \lambda$, the Deodhar down-lift of z_ρ to the coset of λ is z_λ if and only if $D(\lambda) \setminus D(\rho)$ is an antichain.*

Proof outline. Lemma A is proved by induction on the number of boxes, peeling a corner. It uses Deodhar's property Z (Björner–Brenti, Prop. 2.2.7) and the trichotomy for the minimal representative of a shifted shape: a simple reflection adds a box, removes a box, or right-multiplies by W_P. A subword argument shows that the second down-lift starts at z_λ. The two antichains $D(\lambda) \setminus D(\rho)$ and $D(\kappa) \setminus D(\lambda)$ are then $g^{-1}(e)$ and $g^{-1}(o)$. The bijection is forced: two V-maps on θ differ only on isolated boxes of θ (Lemma 4.3), these lie on distinct diagonals, and so distinct V-maps have distinct monomials.

Consequence: a second proof, with different inputs. Remark 4.22 gives a second proof of the formula of Theorem 4.20 from LS12, (KT), Lemma 3.2 and the dictionary $\pi^*[\mathcal{O}_{X_\kappa}]_T = [\mathcal{O}_{z_\kappa}]$, $L^{-1} = L^{-\omega_m}$ of Remark 4.21. It uses (G1) only at the base point (to identify L with L^{ω_m}), and it uses neither (LS), Stanley reciprocity nor [BR12]. (KT) is used only in its spanning form: it puts $\pi^*[\mathcal{O}_{X_\kappa} \otimes L^{-1}]_T$ in the $R(T)$-span of the $[\mathcal{O}_{z_\rho}]$, so the terms of (3.6) with $w \notin W^P$ vanish. The non-equivariant values c_ρ then also need (G0), as at the end of the proof of Theorem 4.20. Remark 4.21 was cold-read on 2026-10-30 and fixed in `ecae68f`: the comparison with LS12 (3.6) is now made in the $R(T)$-basis $\{[\mathcal{O}_w]\}$ of $K_T(G/B)$ using only the formula of Theorem 4.20 (not its uniqueness), and the Schubert dictionary $\pi^*[\mathcal{O}_{X_\kappa}]_T = [\mathcal{O}_{z_\kappa}]$, $z_\kappa = w_0 w_\kappa w_{0,P}$, is stated; the dependency of Remark 4.22 on (KT) is spanning-form only.

What remains open. Question 8.6 (p. 32) now asks only whether this matching is compatible with the bijection of Proposition 3.3 (LS paths of shape $j\omega_m \leftrightarrow$ order-preserving maps G, p. 9).

Status. Proved for all m. Machine checks: Lemma A and the set-level description hold for $m \leq 5$ with 0 failures. An untwisted negative control ($z_\kappa\omega = +v_\kappa$) breaks the claim, with 14/45 failures of the set-level description at $m = 3/4$. The weights agree with the $m \leq 4$ monomial check of Remark 4.21.

4 Erratum to my email of 2026-10-26

In my email of 2026-10-26 (WIP `d118e27`) I wrote "Prop 6.16(a)" for the half-step proposition. It should read **Proposition 6.17(a)** (`prop:T`, $s_{x+1/2} = Ts_x$), and that was already its number at `d118e27`. Remark 6.16 is `rem:serre` (the cohomological formula). Nothing else in that email changes.

5 What I'd like from you

1. A review of the **BR-free proof of Theorem 4.20** (§4.5, pp. 17–20): Lemma 4.15, Proposition 4.16 (the reciprocity step), Proposition 4.17 (the weighted peel), Lemma 4.18 (independence), and the passage from characters to classes via (KT) and (G0). Please also check the dependency claim that Theorem 4.8 needs only (G0), (G1), (LS), (KT) and Stanley reciprocity.

2. A review of **Remark 4.22** (p. 21, about 25 lines), in particular the direction convention and the claim that the second proof uses (G1) only at the base point. The full proof of Lemma A is in the proof file `proofs/2026-10-30-q48-ls-vmap.md`.

This does not replace your standing review of the FPSAC cut `6dbecb8`, and it is less urgent.